

SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **26047**
- Problem Statement Title-**Patient Case-Taking Software**
- Theme- **MedTech / BioTech / HealthTech**
- PS Category- **Software**
- Team ID-
- Team Name- **Team Paradox**



IDEA TITLE

MediKiosk: Patient Case-Taking Software



A) Adaptive Dual-Mode Interview

- Voice + touch interview using SOCRATES/OPQRST; ontology-constrained questions
- Coverage meter, AYUSH Mode (Dashavidha Pariksha) option for traditional assessment



B) Document Digitization Pipeline

- Capture → Quality check → Layout classify → OCR → Entity extraction → Timeline



C) Physician-Ready Structured Summary

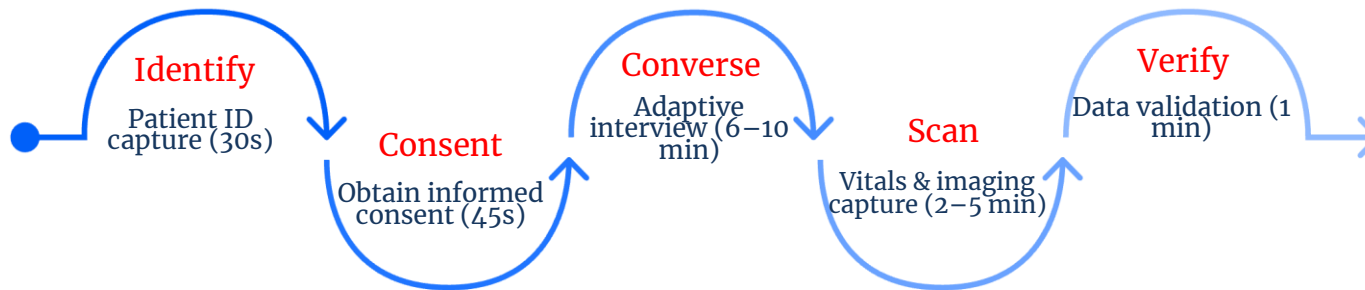
- Editable, source-labelled, confidence-scored, bilingual summary (EN/Hin)



D) Consent & Integration + Break-Glass

- DPDP-2023 consent flows, ABDM-FHIR connectors, emergency break-glass access (differentiator)

5-step Patient Journey (strip)

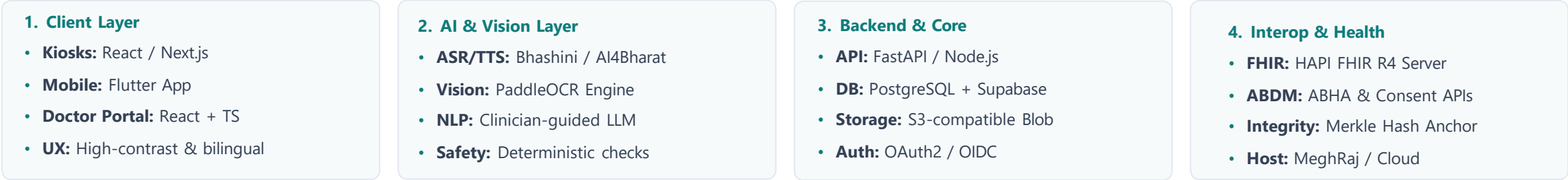


AI DRAFTS. PHYSICIAN VERIFIES. NEVER DIAGNOSES. NEVER PRESCRIBES. — The physician gate is non-negotiable and enforced at system level.

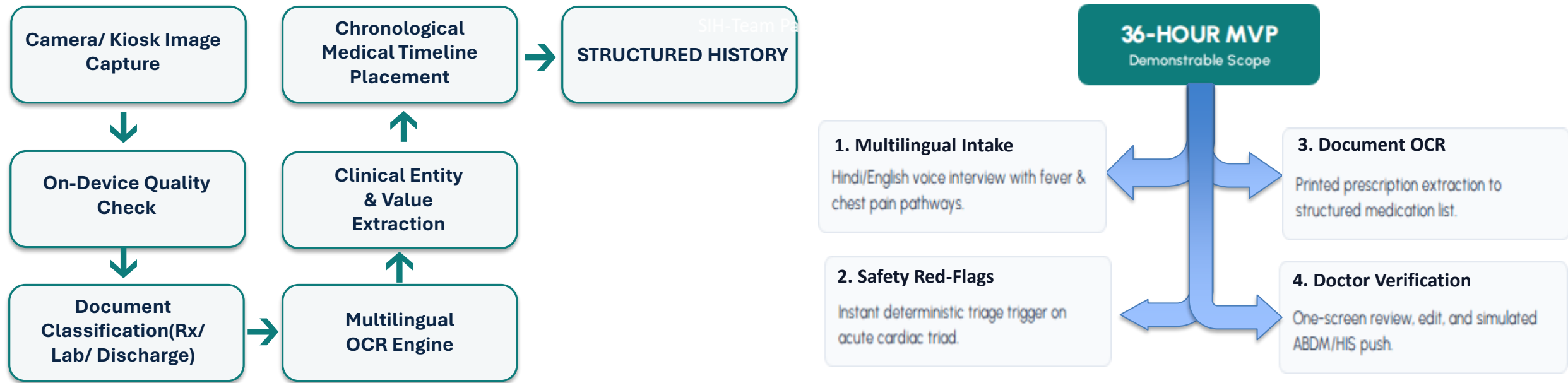
Key Innovations

- Designed for real 2-minute OPD realities: patient-facing, rapid intake
- Voice⊕touch parity for low literacy & noisy settings
- Deterministic clinician-tuned red-flag layer (no black-box alerts)
- Fused speech + paper intake with hash-anchored tamper evidence (no blockchain baggage)

Tech Stack:



Document Pipeline:



HIGH FEASIBILITY:

Mature components + government-backed rails + integration-first

MATURE COMPONENTS

Bhashini / AI4Bharat
PaddleOCR
Constrained dialogue

GOVERNMENT RAILS

ABDM sandbox
HAPI FHIR
DPDP 2023 design

TEAM READINESS

React / TypeScript / Vite
Supabase scaffolding
MediCareX • Clinic Ease

RISK → MITIGATION:

RISK
Handwriting OCR
Noisy ASR / accents
False red-flags
Integration resistance

MITIGATION
Patient verification + confidence + physician gate
Touch fallback + assisted mode + language model selection
Deterministic clinician-tuned rules + human triage
Phased adapters + standalone pilot + no rip-and-replace

HANDWRITING OCR IS THE HARDEST NUT.

CONFIDENCE → PATIENT VERIFY →
PHYSICIAN GATE

MVP ≠ PRODUCTION • MOCK HIS • SYNTHETIC DATA

IMPACT AND BENEFITS

3,28,193

AIIA OPD patients
FY 2024–25

AIIA / Ayush 2025

≈2 min

Average primary-care
consultation, India

BMJ Open 2017

IMPACT LOGIC:

- HIGH PATIENT VOLUME
- + SHORT CONSULTATION
- + HIGH VALUE OF HISTORY
- + PAPER FRICTION
- + ABDM OPPORTUNITY
- ↓ MEDIKIOSK

75.9%

History-alone diagnostic
contribution, pooled

5 studies, 1947–2009

98.6%

Patients maintaining
paper records

IJMR 2024

MEDIKIOSK TARGETS:

30–50%

history-time reduction

8–12 min

intake during wait

>80%

question completion

100%

configured red-flags
routed

MINUTES

break-glass access

DASHAVIDHA

feasible

TARGETS = GOALS, NOT ACHIEVED RESULTS

PATIENT

Speaks in own language; no repetition;
allergies surfaced automatically

DOCTOR

Structured, verified history on screen;
30s review vs. 8 min elicitation

HOSPITAL

Higher OPD throughput; reduced
duplicate tests; ABDM-ready

MINISTRY OF AYUSH

AYUSH-aware clinical payloads on
ABDM rails; population-level data

Problem Evidence:

Hobbs FR et al. "International variations in primary care physician consultation time: a systematic review of 67 countries," *BMJ Open* 2017;7:e017902 (8 Nov 2017)

WHO Fact Sheet — Patient Safety (13 Sep 2023) — diagnostic errors in 5–20% of encounters

Govil D et al. "Medication errors in Indian hospitals: a systematic review," *BMJ Quality & Safety* 2024 (2 Oct 2024) — 2.5× risk from history-taking failure

Indian Journal of Medical Research (IJMR) 2024 — 98.6% on paper records; 27% have lost them

History-Taking Yield:

Platt FW. *The Lancet* 1947 — 74% diagnostic yield from history

Hampton JR et al. *BMJ* 1975 — 82.5% diagnostic yield from history

Pooled mean (5 studies, 1947–2009): 75.9% (95% CI 67.9–83.9)

Policy & Infrastructure:

PIB, 30 May 2026 — 90 Crore ABHA IDs issued to date

PIB, 11 Feb 2025 — Under 1.6 lakh ABDM-enabled facilities nationwide

Ministry of Ayush, 16 Sep 2025 — "11 Years of AYUSH" report; AIIA recorded 3,28,193 OPD patients in FY 2024–25

MoRTH 2024 — 68,584 hit-and-run accidents; thousands of "blank patients" with no accessible history

Regulatory Framework:

DPDP Act 2023 Consent, data minimization, and erasure rights — basis for all data handling in MediKiosk	ABDM Consent Manager Purpose-limited sharing with revocable, patient-controlled consent artefacts
HL7 FHIR Standard Interoperability layer for structured health information exchange across facilities	